Aluminium Alloys

Extraction of Aluminium

- Aluminium ore (Al2O3), or bauxite has its aluminium extracted via the Bayer process (type of electrolysis)
- The process involved combining bauxite ore and cryolite (Al3F2) to lower the melting point of bauxite, before being electrolysed to extract the molten aluminium, before being drained.

Properties of (pure) Aluminium

- Aluminium has a low density of 2.7g/cm^3, one of the lightest metals, ~1/3 the density of copper
- Al also has a high electrical conductivity, and with a better thermal conductivity than steel.
- Al has 2/3 of Cu's conductivity, making it a better conductor than Cu on per weight basis.
- Aluminium also has a high corrosion resistance, due to the dense aluminium oxide layer on its surface(oxidised Al) However, the layer is thin and can be improved via anodising (further oxidising the aluminium via electrolysis)
- Aluminium is also very ductile and malleable, making it useful to be formed into thin sheets, for food packaging.
- Al also has a high reflectivity for visible and infrared light, making it useful for reflectors.
- When finely divided, aluminium is impervious to water (aka does not allow water to pass through) making it useful for paints to provide a coating.
- However, Al has a low tensile strength of 90N/mm^2, however can be improved via cold working, alloying, and heat treatment (after alloying).

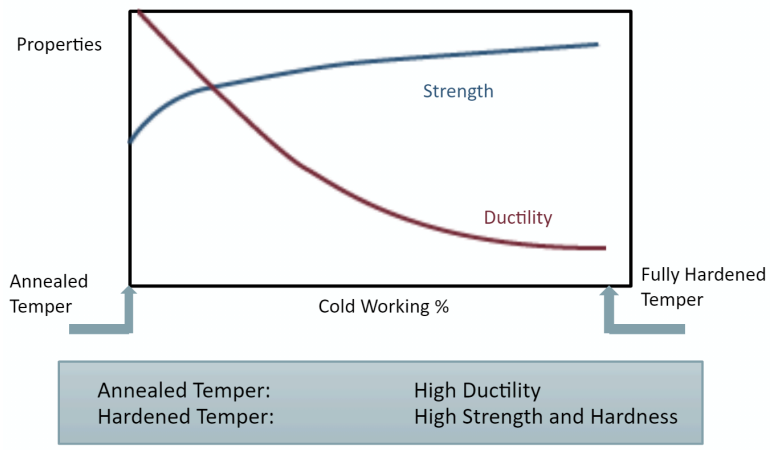
Aluminium Alloys

- Al alloys can be divided by how they are shaped, into wrought and cast Al, and further divided into whether it can be heat treated or not.
- Wrought Aluminium
  - Wrought Aluminium refers to alloys shaped by working, like forging, rolling, or extruding. These alloys have to be very ductile and malleable.

Recall that the more cold worked a metal is, the strength increases and the ductility decreases.

Two allows of different amount of cold working are said to have different tempers.

The two main ones are the annealed temper (minimal cold working done), with high ductility, and the fully hardened temper (maximum cold working done), with high strength and hardness.



Cast Aluminium

Cast aluminium refers to alloys shaped by casting, like sand casting (using charged sand as a mould for the molten alloy) or pressure die casting (injecting the molten alloy into the mould at high pressure)

These cast alloys have to flow well when molten (good fluidity/low viscosity), low shrinkage (does not shrink under state change), and a short freezing range (to ensure the metal cools immediately after injected into mould, so the molten metal does not damage the shape)

Classification

Wrought Aluminium

Wrought Al is classified by a 4 digit number.

Wrought Aluminium Alloy	Alloy Designation
Aluminium(99% or more)	1xxx
Major Alloying Element:	
Copper	2xxx
Manganese	3xxx
Silicon	4xxx
Magnesium	5xxx
Magnesium and Silicon	6xxx
Zinc	7xxx

Heat Treatable

Non-Heat Treatable

❖ Strength and hardness of the alloy can be improved by cold working

1xxx refers to (practically) pure aluminium while other numbers have one major alloying element.

Some alloys like AlCu, AlMgSi and AlZn are heat treatable, while others aren't.

Different alloying elements will affect their strength, ductility, and their applications:

Alloy	Strength	Ductility	Applications
2xxx AlCu	high	low-moderate	aircraft due to high strength to weight ratio
3xxx AlMn	moderate	high	corrogated roofs, window frames
4xxx	moderate	modertae(actually idk lol)	Forged piston
5xxx AlMg	moderate-high (best non heat treatable strength)	high	train/bus bodies
6xxx AlMgSi	moderate-high	moderate-high	Extruded profiles, scaffolds,
7xxx AlZn	high	low	aircraft

One special aluminium alloy is Duralumin AA2024,

Cast Aluminium

Cast Al are classified by 3 digits and 1 decimal position system.

2xx.x alloys can maintain its strength at high temperatures, making it useful in engines.

Major Alloying Element	Alloy Designation	Typical Application
Copper	2xx.x	Y-alloy. Engine piston cylinder head in combustion engine
Silicon	4xx.x	Use in 'as- cast' Oil sumps, gear box and radiators

The first number represents the main alloying element, the same as wrought Al alloys.

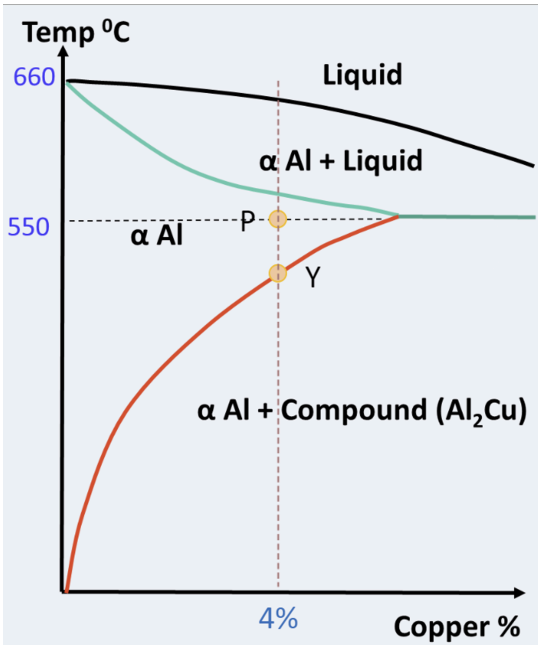
4xx.x alloys can be used as cast (without any post treatment)

Note that cast alloys are often not used for structural components due to their low strength compared to wrought alloys.

Heat Treatment of Aluminium Alloys

There are 2 steps to heat treat aluminium alloys, solution treatment followed by precipitation hardening/Aging. The main focus will be on alloys with Al and Cu.

Phase Diagram of Al-Cu/Slow cooling of Al-Cu Alloy



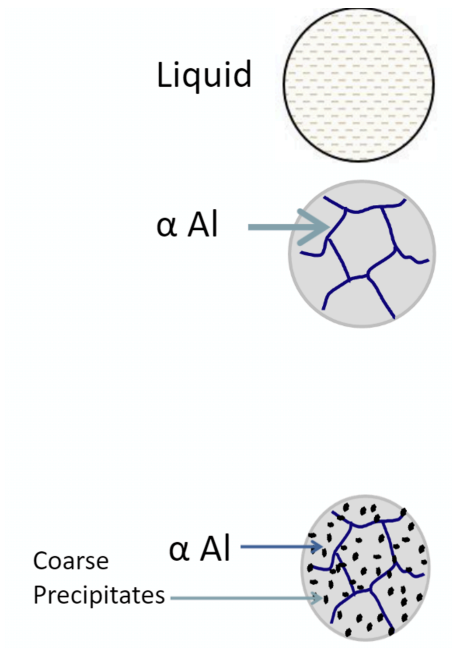
The phase diagram shows different phases of liquid alloy, alpha aluminum (a solid solution of FCC aluminium with copper dissolved), and Al2Cu precipitate.

Consider the cooling of duralumin. As the alloy cools to 550C, the microstructure changes from liquid to solid grains of alpha Al. As it further cools to the red line, it will reach the maximum amount of copper dissolved in the Al at that temperature (aka if point Y has a temperature T, it cannot move rightwards to increase the copper dissolved (as the copper will no longer be properly dissolved in the aluminium), thus Y has max aluminium)

As it further cools, solubility decreases and less Cu is dissolved in the Al, causing coarse precipitates of Cu to form.

Thus, at room temperature the solubility of copper in aluminium will be minimal, ~0, while **at 550C, the solubility will be at a maximum of ~5%**

The final microstructure at room temperature will be soft and weak.

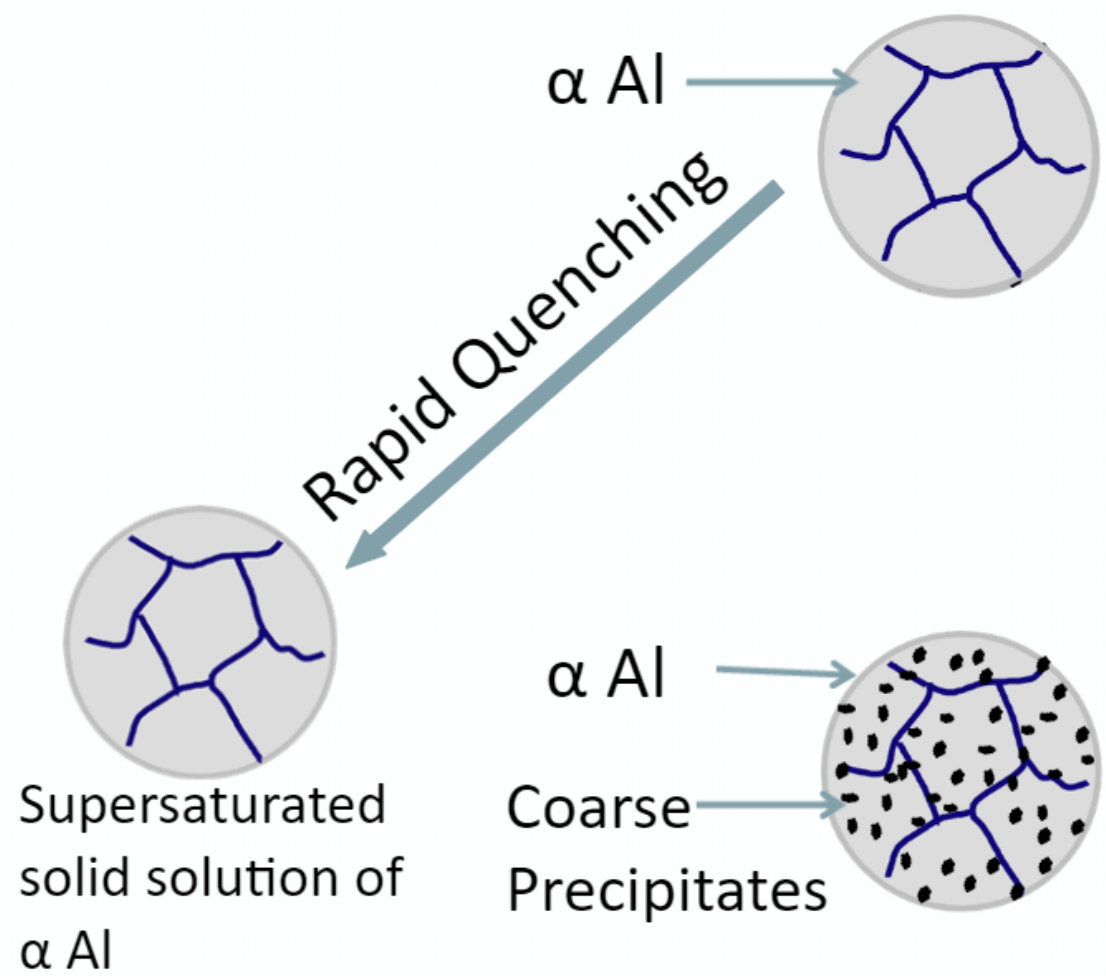


Solution Treatment

Solution treatment has a similar procedure to the hardening of plain carbon steels.

The alloy is heated to 550C and held for a sufficient amount of time to dissolve the precipitates of Cu, obtaining a microstructure of only alpha Al, before being rapidly quenched in water or oil to preserve the microstructure and prevent the formation of Cu precipitates.

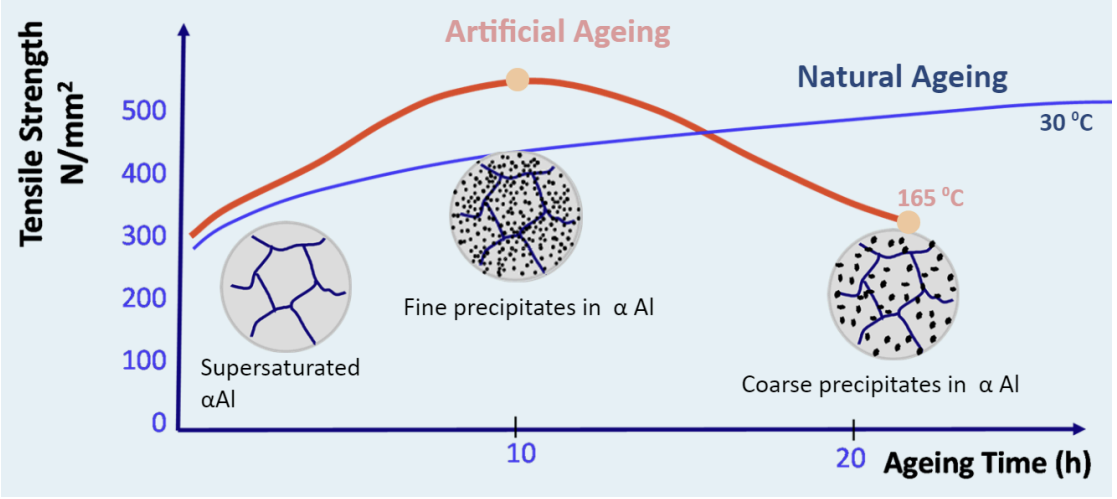
The final microstructure is now a super saturated solid solution of alpha Al, which is very ductile, which allows it to be worked, before undergoing precipitation hardening



Precipitation Hardening/Ageing

Ageing is a process that produces precipitates in the alloy, hardening it, by holding the alloy at a certain temperature for an amount of time.

There are 2 ways to age the alloy, either with Natural Ageing or Artificial Ageing.



Natural Ageing

The component is held at room temperature(~30C) to slowly develop strength and hardness over 4 days. However, this method is very slow.

Artificial Ageing

The component is heated to 165C for 10 hours, before being cooled in air. This method is much faster than natural ageing.

From the graph, we can see that 10hours gives us the peak tensile strength for ageing, as too little or too much holding time will not give us the maximum tensile strength.

As the component is held at 165C, the tensile strength increases as fine precipitates form in the aluminium. However, past 10hours, the precipitates become more coarse, causing a decrease in tensile strength and an increase in ductility. This aluminum alloy is considered overaged.